Category-Aware Location Embedding for Point-of-Interest Recommendation

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Abstract

We propose a novel neural model that generates a POI embedding incorporating sequential and categorical information from POIs. Our model consists of a check-in module and a category module.

Motivations

- With the advent of Location-based Social Networks, users can share check-in data using their mobile devices, together with contextual information
- Successful results of considering Pre-Training models

Objectives

- Propose a POI embedding model, focusing on the contextual information
- Account sequential and categorical information
- Examine the effect of contextual information in pre-training model of POI

Proposed Method

- We propose a category-aware POI embedding model called CATAPE
- That captures both the geographical and categorical information of POIs
- Check-in Module: Visited POIs by users are geographically influenced by each other.

$$\phi_{ch}(l, S_l) = arg\ max \prod_{l \in L} [\prod_{l_w \in W(l,S_l)} p(l_w|l)] \quad (1)$$

- Category Module: Categories of POIs visited show the user's interests.

$$\phi_{ct}(l, c) = arg\ max \prod_{l_i \in L} \prod_{c_j \in C_i} \prod_{c_q \in Q(c_j)} p(c_q|c_j, l_i) \quad (2)$$

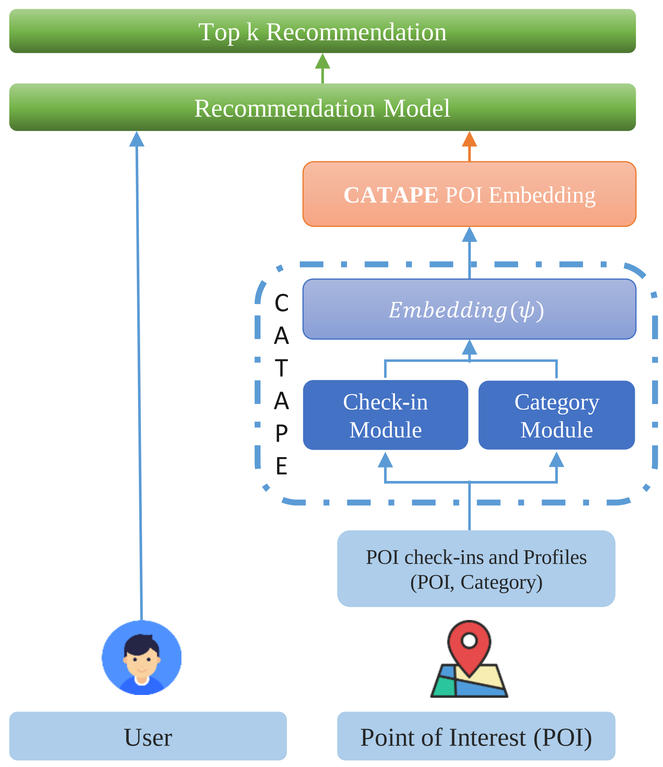
- Objective Function

$$\psi_{emb} = \phi_{ch}(l, S_l) + \phi_{ct}(l, c) \quad (3)$$

Model Workflow (CATAPE)

Figure 1 shows the workflow of pre-trained POI embedding from CATAPE to produce top k recommendation list.
CATEPE learns high-dimensional POI embeddings. Then, the learned POI embedding is fed to a recommender model which is able to utilize this information to provide accurate recommendation.

Figure 1 – CATAPE Model Overview



Experiments

- Datasets: Yelp and Gowalla
  - **Yelp:** 357,410 check-ins made by 6,254 users on 21,240 POIs and 507 unique categories.
  - **Gowalla:** 456,988 check-ins made by 10,162 users at 24,250 POIs with 417 categories.
- Model variations:
  - CATAPE-NoCat
  - CATAPE

Results

Table 1 – Performance comparison with baselines in terms of Precision@k and Recall@k for $k \in \{5,10,20\}$ on Gowalla and Yelp. The superscripts † and ‡ denote significant improvements compared to baselines and CATAPE-NoCat, respectively ($p < 0.05$).

Method	Yelp						Gowalla					
	P@5	P@10	P@20	R@5	R@10	R@20	P@5	P@10	P@20	R@5	R@10	R@20
USG	0.0282	0.0244	0.0197	0.0281	0.0523	0.0753	0.0502	0.0471	0.0413	0.0517	0.0568	0.0625
MGMPFM	0.0197	0.0173	0.0136	0.0211	0.0293	0.0493	0.0281	0.0215	0.0197	0.0263	0.0291	0.0319
BPRFM	0.0285	0.0221	0.0185	0.0296	0.0361	0.0599	0.0493	0.0443	0.0342	0.0497	0.0529	0.0581
RankGeoFM	0.0421	0.0362	0.0292	0.0392	0.0673	0.0838	0.0567	0.0501	0.0492	0.0591	0.0642	0.0718
HGMF	0.0532	0.0491	0.0401	0.0478	0.0702	0.0915	0.0798	0.0711	0.0683	0.0715	0.0773	0.0819
Metric Factorization	0.0593	0.0552	0.0481	0.0533	0.0782	0.0974	0.0821	0.0782	0.0717	0.0784	0.0814	0.0862
CATAPE-NoCat	0.0641†	0.0613†	0.0568†	0.0589	0.0831†	0.1013†	0.0892†	0.0828†	0.0784†	0.0815†	0.0898†	0.0979†
CATAPE	**0.0702**†‡	**0.0692**†‡	**0.0631**†‡	**0.0621**†‡	**0.0881**†	**0.1121**†‡	**0.0924**†‡	**0.0894**†‡	**0.0813**†‡	**0.0872**†	**0.0953**†‡	**0.1283**†‡

Conclusion

We introduced a novel POI embedding model and demonstrated the importance of characteristics of POIs in POI embedding. Our model captures the sequential influence of POIs from check-in sequence of users, as well as, characteristics of POIs using the category information.

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